Genetics: From Genes to Genomes

Fifth Edition

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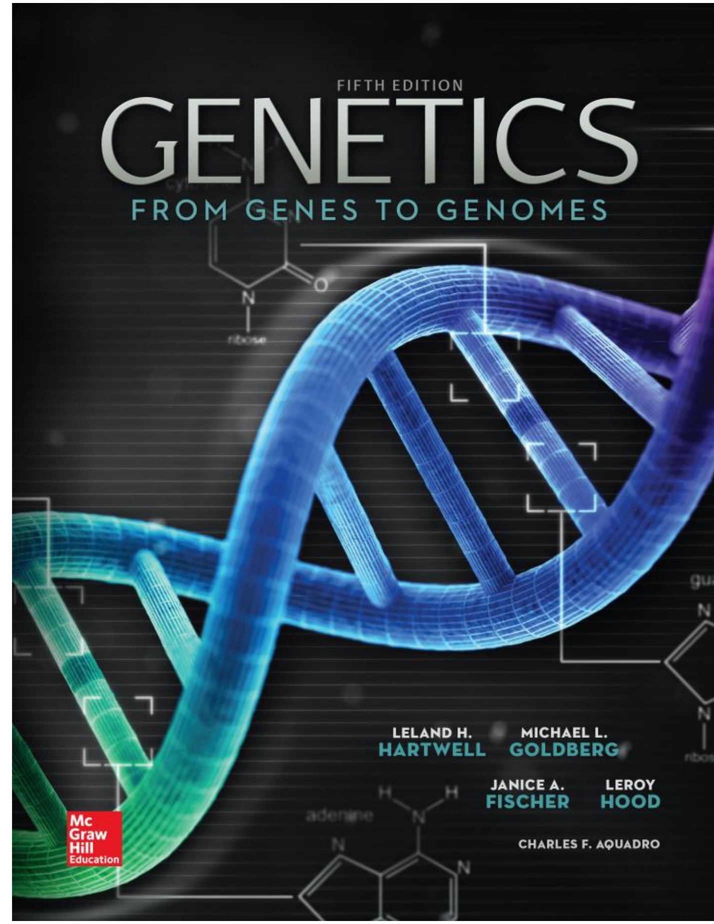
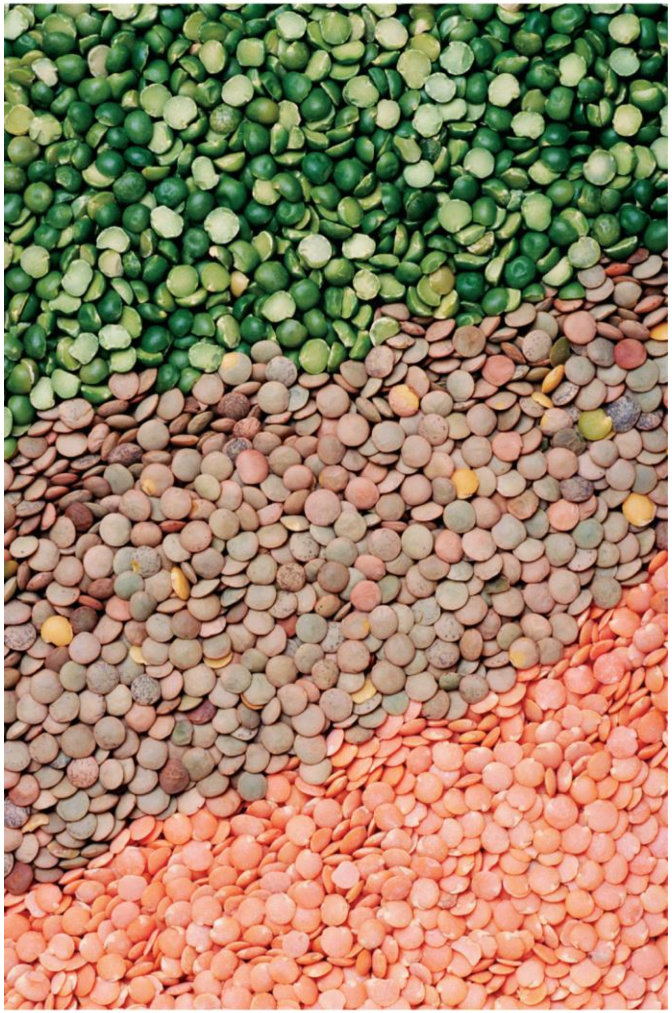


Extensions to Mendel's laws



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CHAPTER OUTLINE

- 3.1 Extensions to Mendel for Single-Gene Inheritance
- 3.2 Extensions to Mendel for Multifactorial Inheritance

Some phenotypic variation poses a challenge to Mendelian analysis

Treasure your exceptions…They tell there is more to come and where the next construction is to be. –William Bateson

Crosses of pure-breeding lines can result in progeny phenotypes that don't appear to follow Mendel's rules

- **No definitively dominant or recessive allele**
- **More than two alleles exist**
- **Multiple genes involved**
- **Gene-environment interactions**

Lentils come in an array of colors and patterns

3.1 Extensions to Mendel for single-gene inheritance

Learning Objectives:

- Categorize allele interactions as completely dominant, incompletely dominant, or codominant.
- Recognize progeny ratios that imply the existence of recessive lethal alleles.
- Predict from the results of crosses whether a gene is polymorphic or monomorphic in a population.

Extensions to Mendel for single-gene inheritance

1. Dominance is not always complete
   - Incomplete dominance
   - Codominance
2. A gene may have >2 alleles
3. Pleiotropy - one gene may contribute to several characteristics
   - Recessive lethal alleles
   - Delayed lethality

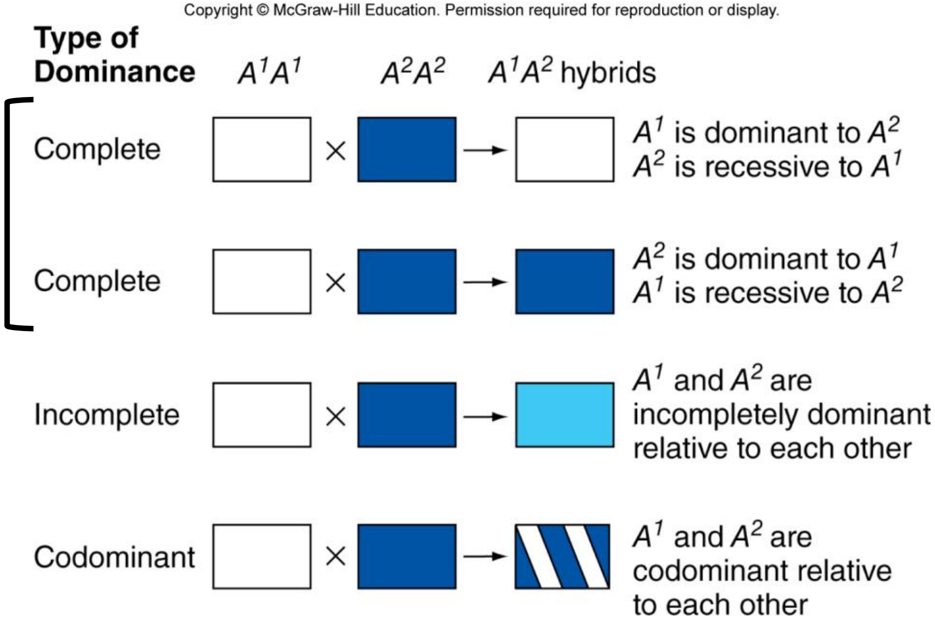
Summary of different dominance relationships

The phenotype of the heterozygote defines the dominance relationship of two alleles

Complete dominance: Hybrid resembles one of the two parents

Incomplete dominance: Hybrid resembles neither parent

Codominance: Hybrid shows traits from both parents



Flower color in snapdragons is an example of incomplete dominance

Crosses of pure-breeding red with pure-breeding white results in all pink F_1 progeny

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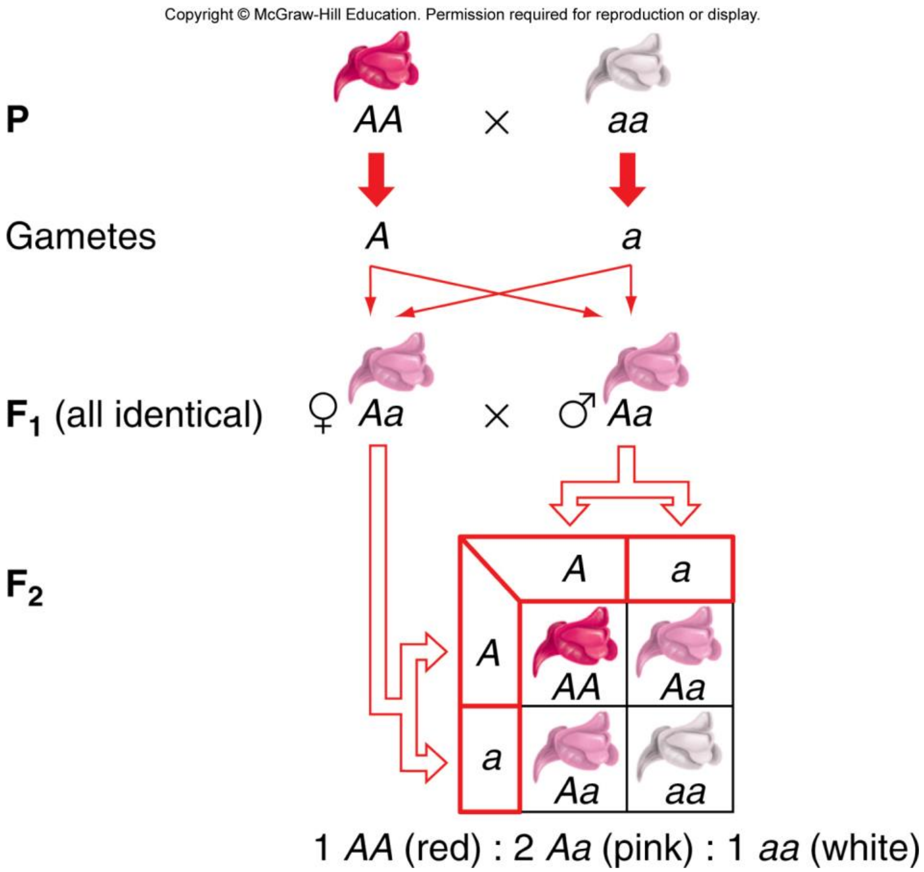
Pink flowers in snapdragons are the result of incomplete dominance

F_2 progeny ratios:

- 1 red (*AA*)
- 2 pink (*Aa*)
- 1 white (*aa*)

Phenotype ratios reflect the genotype ratios



Molecular explanation for incomplete dominance in snapdragon flower color

The normal allele produces an enzyme involved in pigment production

- **AA = 2 normal alleles = red color**
- **Aa = 1 normal allele = pink color (less enzyme)**
- **Aa = 0 normal alleles = white color (no functional enzyme)**

In codominance, the F_1 hybrids display traits of both parents

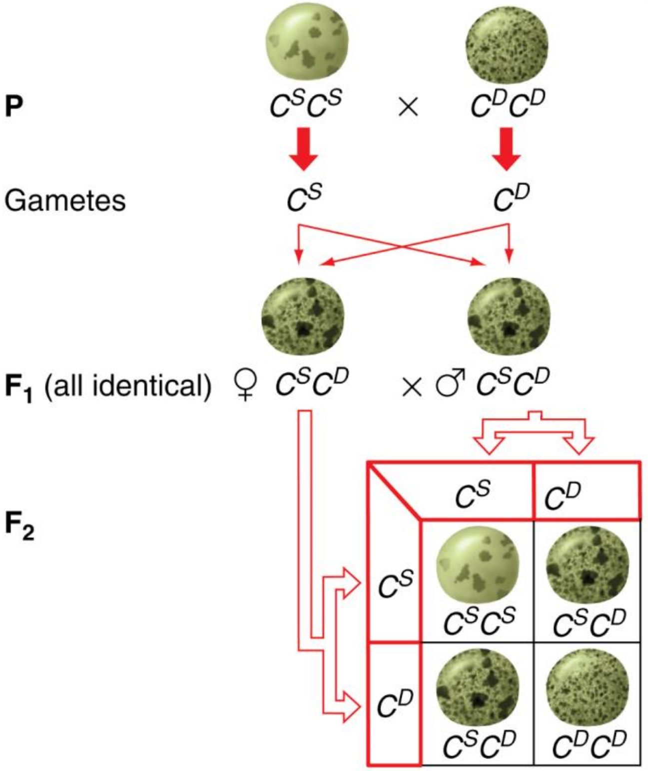
Spotted (*C^SC^S*) x dotted (*C^DC^D*)

<u>All</u> F_1 progeny are spotted <u>and</u> dotted (*C^SC^D*)

F_2 progeny ratios:

- **1 spotted (*C^SC^S*)**
- **2 spotted and dotted (*C^SC^D*)**
- **1 dotted (*C^DC^D*)**

Phenotype ratios reflect the genotype ratios



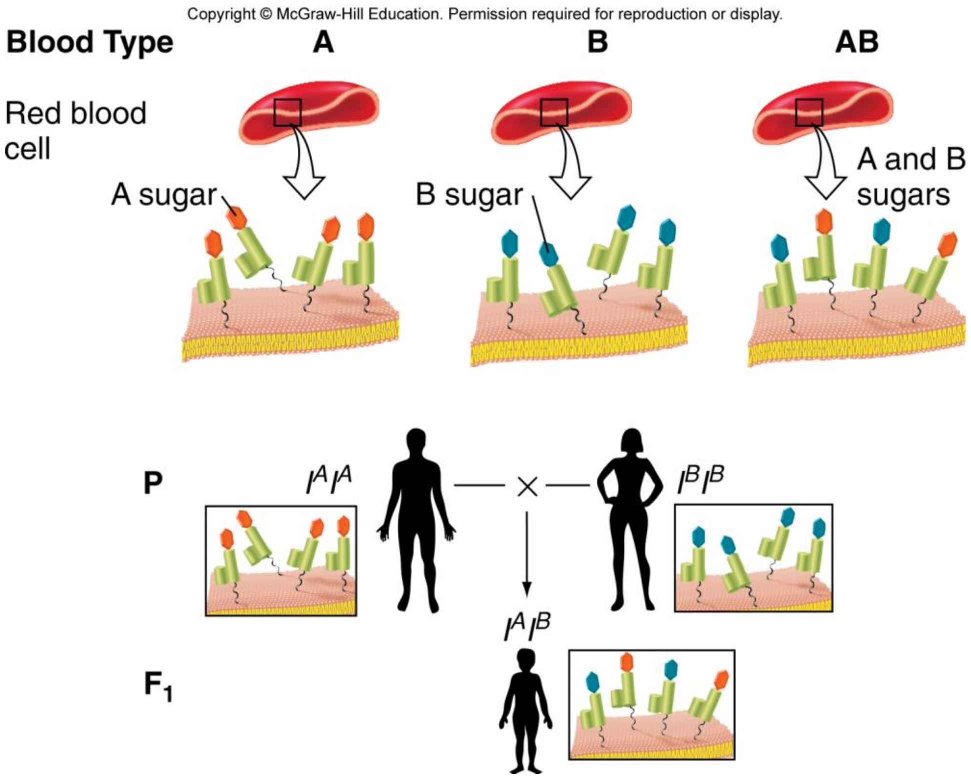
1 C^SC^S (spotted) : 2 C^SC^D (spotted/dotted) : 1 C^DC^D (dotted)

In codominance, the F_1 hybrids display traits of both parents

Gene *I* controls the type of sugar polymer on surface of RBCs

Two alleles, I^A and I^B, result in different sugars

- **$I^A I^A$ individuals have A sugar**
- **$I^B I^B$ individuals have B sugar**
- **$I^A I^B$ individuals have both A and B sugars**



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Dominance relations between alleles do not affect transmission of alleles

Type of dominance (complete, incomplete dominance, codominance) depends on the type of proteins encoded and the biochemical functions of the proteins

Variation in dominance relations do not negate Mendel's laws of segregation

Interpretation of phenotype/genotype relations is more complex

A gene can have more than two alleles

Multiple alleles of a gene can segregate in populations

Each individual can carry only two alleles

Dominance relations are always relative to a second allele and are unique to a pair of alleles

ABO blood types in humans are determined by three alleles of one gene

I^A allele → A type sugar

I^B allele → B type sugar

i allele → no sugar

Genotypes	Corresponding Phenotypes: Type(s) of Molecule on Cell
$I^A I^A$ $I^A i$	A
$I^B I^B$ $I^B i$	B
$I^A I^B$	AB
$i\,i$	O

Six genotypes produce four blood phenotypes

Dominance relations are relative to a second allele

- I^A and I^B are codominant
- I^A and I^B are dominant to i

Medical and legal implications of ABO blood group genetics

Antibodies are made against type A and type B sugars

- **Successful blood transfusions occur only with matching blood types**
- **Type AB are universal recipients, type O are universal donors**

Blood Type	Antibodies in Serum
A	Antibodies against B
B	Antibodies against A
AB	No antibodies against A or B
O	Antibodies against A and B

Blood Type of Recipient	Donor Blood Type (Red Cells) A	B	AB	O
A	+	–	–	+
B	–	+	–	+
AB	+	+	+	+
O	–	–	–	+

Seed coat patterns in lentils are determined by a gene with five alleles

Five alleles for *C* gene: spotted (C^S), dotted (C^D), clear (C^C), marbled-1 (C^{M1}), and marbled-2 (C^{M2})

Reciprocal crosses between pairs of pure-breeding lines is used to determine dominance relations (see Fig 3.6)

Dominance relations are established between pairs of alleles

Three examples from Figure 3.6

1. **marbled-1 ($C^{M1}C^{M1}$) x clear (C^CC^C) → all F_1 marbled-1 ($C^{M1}C^C$)**
   F_2 progeny: 798 marbled-1 (C^{M1}—) and 296 clear (C^CC^C)

2. **marbled-2 ($C^{M2}C^{M2}$) x clear (C^CC^C) → all F_1 marbled-2 ($C^{M2}C^C$)**
   F_2 progeny: 123 marbled-1 (C^{M2}—) and 46 clear (C^CC^C)

3. **marbled-1 ($C^{M1}C^{M1}$) x marbled-2 ($C^{M2}C^{M2}$) → all F_1 marbled-1**
   F_2 progeny: 272 marbled-1 (C^{M1}—) and 72 marbled-2 ($C^{M2}C^{M2}$)

3:1 ratio in each cross indicates that different alleles of the same gene are involved

Dominance series: $C^{M1} > C^{M2} > C^C$

Human histocompatibility antigens are an extreme example of multiple alleles

Three major genes (*HLA-A*, *HLA-B*, and *HLA-C*) encode histocompatibility antigens

- **Cell surface molecules present on all cells except RBCs and sperm**
- **Facilitates proper immune response to foreign antigens (e.g. virus or bacteria)**

Each gene has 400-to-1200 alleles each

- **Each allele is codominant to every other allele**
- **Every genotype produces a distinct phenotype**
- **Enormous phenotypic variation**

Mutations are the source of new alleles

Chance alterations of genetic material arise spontaneously

If mutations occur in gamete-producing cells, they can be transmitted to offspring

Frequency of gametes with mutations is 10^{-4}-10^{-6}

Mutations that result in phenotypic variants can be used by geneticists to follow gene transmission

Nomenclature for alleles in populations

Allele frequency is the percentage of the total number of gene copies for one allele in a population

Most common allele is usually the wild-type (+) allele

Rare allele is considered a mutant allele

Gene w/ only one common wild-type allele is **monomorphic**

Gene w/ more than one common allele is **polymorphic**

- High-frequency alleles of polymorphic genes are **common variants**

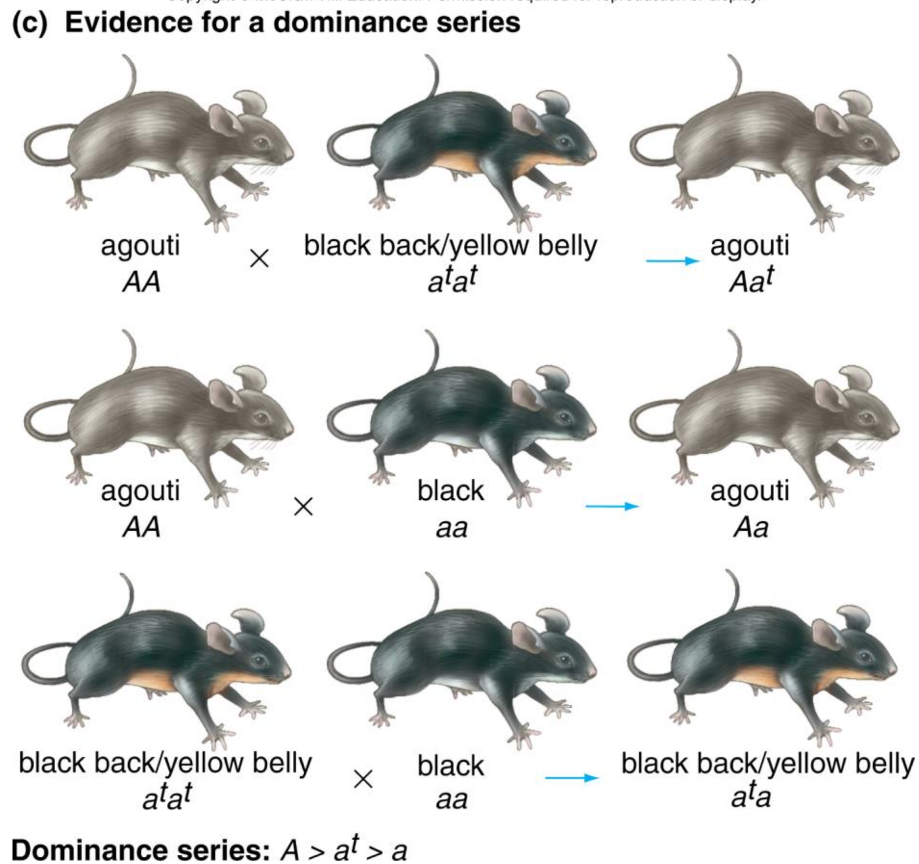
The mouse *agouti* gene controls hair color: One wild-type allele, many mutant alleles

Wild-type agouti allele (*A*) produces yellow and black pigment in hair

14 different *agouti* alleles in lab mice, but only *A* allele in wild mice

e.g. mutant alleles *a* and a^t

- ***a* recessive to *A***
  - ***aa* has black only**
- **a^t dominant to *a* but recessive to *A***
  - **a^ta^t mouse has black on back and yellow on belly**



One gene may contribute to several characteristics

Pleiotropy is the phenomenon of a single gene determining several distinct and seemingly unrelated characteristics

- **Many aboriginal Maori men have respiratory problems and are sterile**
- **Defects due to mutations in a gene required for functions of cilia (failure to clear lungs) and flagella (immotile sperm)**

With some pleiotropic genes

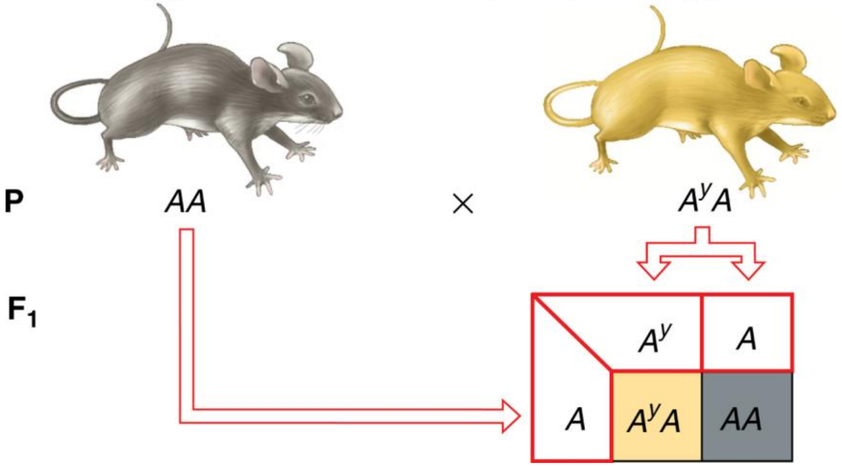
- **Heterozygotes can have a visible phenotype**
- **Homozygotes can be lethal (e.g. AY allele of agouti gene in mice, see Fig 3.9)**

The A^Y allele produces a dominant coat color phenotype in mice

A^Y allele of *agouti* gene causes yellow hairs

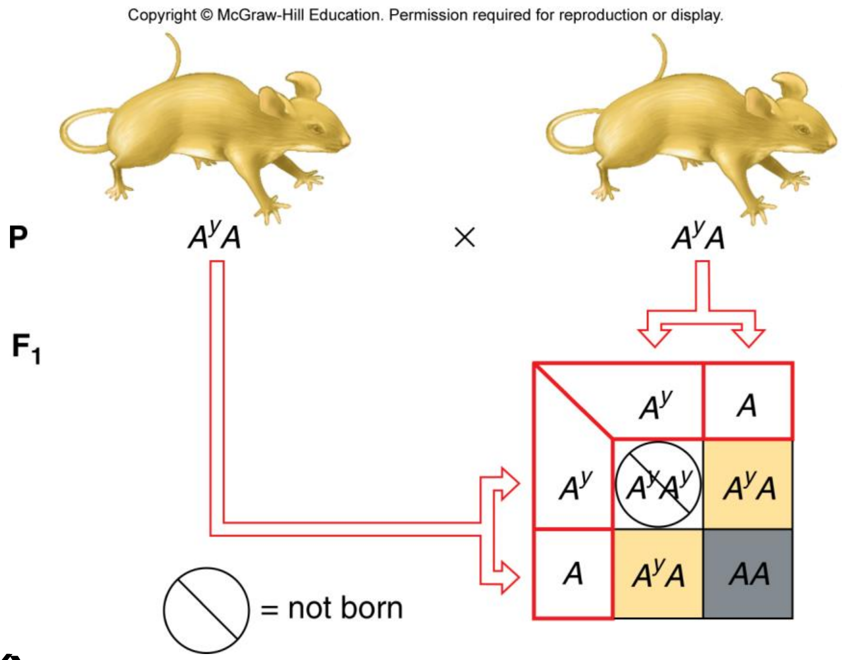
Cross agouti x yellow mice

- Progeny in 1:1 ratio of agouti to yellow
- Yellow mice must be heterozygous for A and A^Y
- A^Y is dominant to A in determining hair color



The A^Y allele is a recessive lethal allele

***A^Y* is <u>dominant</u> to *A* for hair color, but is <u>recessive</u> to *A* for lethality**



Cross yellow x yellow mice

- **F_1 mice are 2/3 yellow and 1/3 agouti**

2:1 ratio is indicative of a recessive lethal allele

- **Pure-breeding yellow (A^YA^Y) mice cannot be obtained because they are not viable**

Extensions to Mendel's analysis explain alterations of the 3:1 monohybrid ratio

TABLE 3.1 For Traits Determined by One Gene: Alterations of the 3:1 Monohybrid Ratio

What Mendel Described	Extension	Extension's Effect on Heterozygous Phenotype	Extension's Effect on Ratios Resulting from an $F_1 \times F_1$ Cross
Complete dominance	Incomplete dominance Codominance	Unlike either homozygote	Phenotypes coincide with genotypes in a ratio of 1:2:1
Two alleles	Multiple alleles	Multiplicity of phenotypes	A series of 3:1 ratios
All alleles are equally viable	Recessive lethal alleles	Heterozygotes survive but may have visible phenotypes	2:1 instead of 3:1
One gene determines one trait	Pleiotropy: One gene influences several traits	Several traits affected in different ways, depending on dominance relations	Different ratios, depending on dominance relations for each affected trait

A comprehensive example: Sickle-cell disease

Hemoglobin transports oxygen in RBCs

- **Two subunits – alpha (α) globin and beta (β) globin**

The β-globin gene has multiple alleles

- **Normal allele ($Hb\beta^{A}$) and 400 mutant alleles.**
- **Most common mutation of β-globin (HbβS) causes sickle-cell disease**

Pleiotropic – affects >1 trait (deformed RBCs, anemia, heart failure, resistance to malaria)

- **Recessive lethality – heart failure**

Different dominance relations for different phenotypic aspects of sickle-cell disease (see Figure 3.10)

Pleiotropy of sickle-cell anemia: Dominance relations vary with the phenotype under consideration

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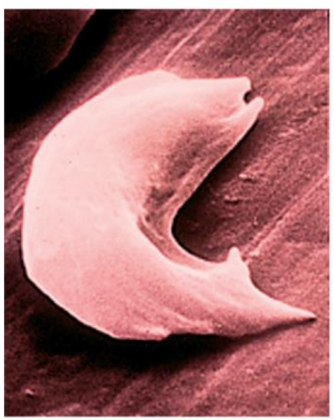

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Phenotypes at Different Levels of Analysis	Normal $Hb\beta^A\ Hb\beta^A$	Carrier $Hb\beta^A\ Hb\beta^S$	Diseased $Hb\beta^S\ Hb\beta^S$	Dominance Relations at Each Level of Analysis
Red blood cell shape at sea level	Normal	Normal	Sickled cells present	$Hb\beta^A$ is dominant $Hb\beta^S$ is recessive
Red blood cell concentration at sea level	Normal	Normal	Lower	
β-globin polypeptide production				$Hb\beta^A$ and $Hb\beta^S$ are codominant
Red blood cell shape at high altitudes	Normal	Sickled cells present	Severe sickling	
Red blood cell concentration at high altitudes	Normal	Lower	Very low, anemia	$Hb\beta^A$ and $Hb\beta^S$ show incomplete dominance
Susceptibility to malaria	Normal susceptibility	Resistant	Resistant	$Hb\beta^S$ is dominant $Hb\beta^A$ is recessive

3.2 Extensions to Mendel for multifactorial inheritance

Learning Objectives:

- Conclude from the results of crosses whether a single gene or multiple genes control a trait.
- Infer the existence of gene interactions, including complementation, epistasis, and redundancy.
- Recognize when a given genotype in different individuals does not correspond to the same phenotype.
- Explain how continuous traits are controlled by multiple alleles of multiple genes.

Extensions to Mendel for multifactorial inheritance

Two genes can interact to determine one trait

- **Novel phenotypes can result from gene interactions.**
- **Fewer phenotypes maybe observed due to complementary gene action.**
- **Epistasis is when an allele at one gene masks the phenotype of alleles at another gene.**
- **Genes that perform the same function are redundant.**

In all of these cases, F_2 phenotypes from dihybrid crosses vary from the expected 9:3:3:1 ratio.

Novel phenotypes resulting from gene interactions, e.g. seed coat in lentils

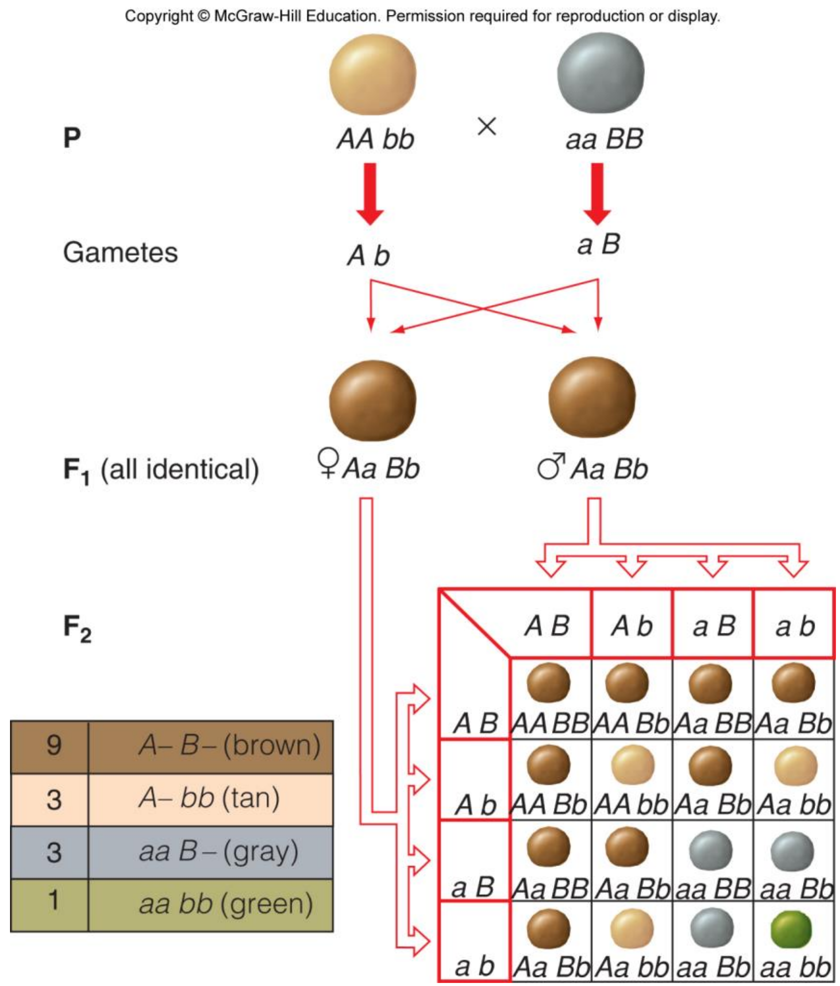
Dihybrid cross of lentils, tan x gray

<u>All</u> F_1 seeds are brown

F_2 progeny:

- **9/16 brown**
- **3/16 tan**
- **3/16 gray**
- **1/16 green**

9:3:3:1 ratio in F_2 suggests two independently assorting genes for seed coat color



Hartwell et al., 5th ed., Chapter 3

Results of self-crosses of F_2 lentils supports the two-gene hypothesis

Phenotypes of F_2 Individual	Observed F_3 Phenotypes	Expected Proportion of F_2 Population*
Green	Green	1/16
Tan	Tan	1/16
Tan	Tan, green	2/16
Gray	Gray, green	2/16
Gray	Gray	1/16
Brown	Brown	1/16
Brown	Brown, tan	2/16
Brown	Brown, gray	2/16
Brown	Brown, gray, tan, green	4/16

***This 1: 1: 2: 2: 1: 1: 2: 2: 4 F_2 genotypic ratio corresponds to a 9 brown: 3 tan: 3 gray: 1 green F_2 phenotypic ratio**

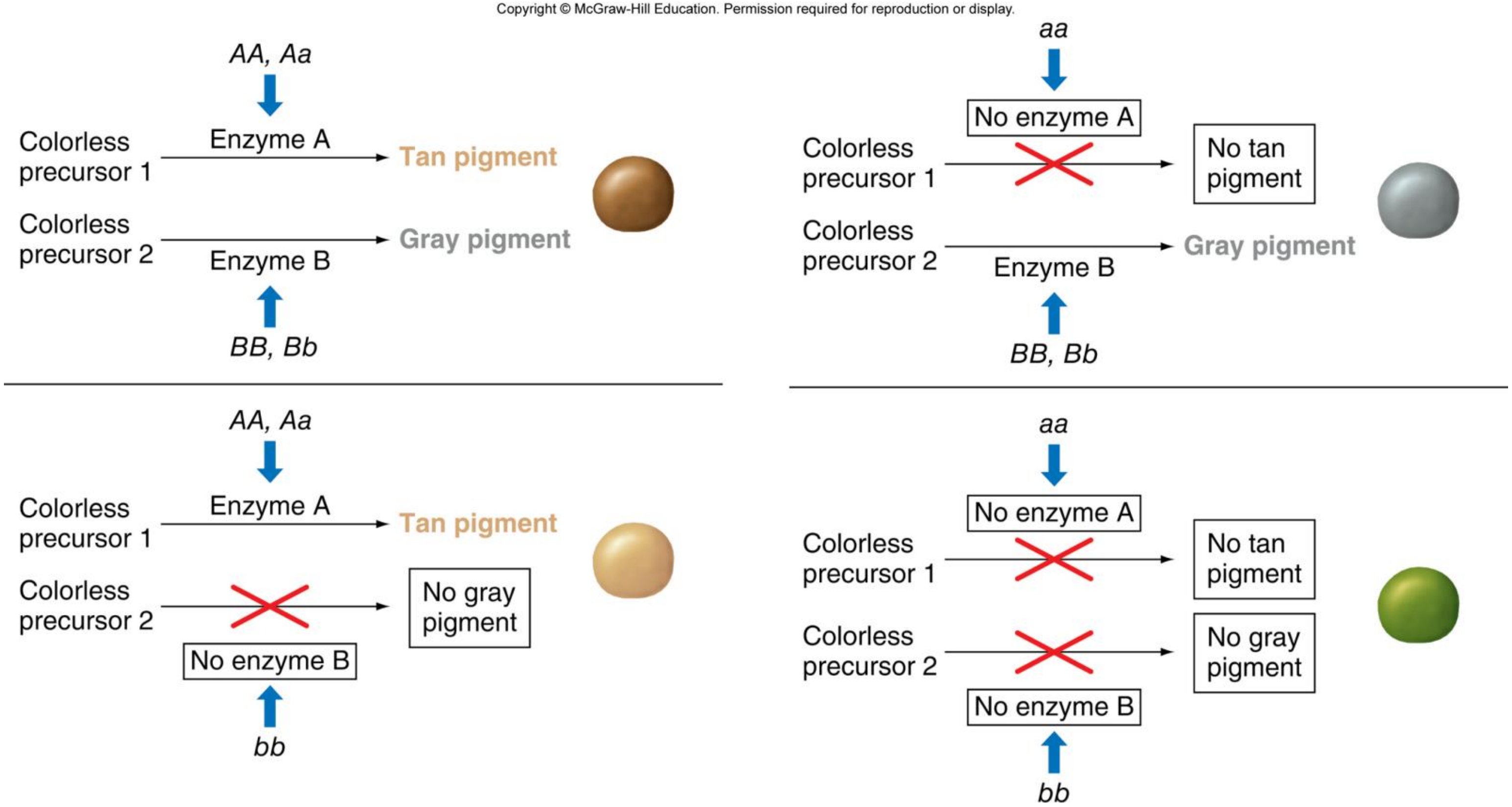
Sorting out the dominance relations by select crosses of lentils

F_2 phenotypes from dihybrid crosses will be in 9:3:3:1 ratio <u>only</u> when dominance of alleles at both genes is complete

Seed Coat Color of Pure-Breeding Parents	F_2 Phenotypes and Frequencies	Ratio
Tan × green	231 tan, 85 green	3 : 1
Gray × green	2586 gray, 867 green	3 : 1
Brown × gray	964 brown, 312 gray	3 : 1
Brown × tan	255 brown, 76 tan	3 : 1
Brown × green	57 brown, 18 gray, 13 tan, 4 green	9 : 3 : 3 : 1

A biochemical model for inheritance of lentil seed color

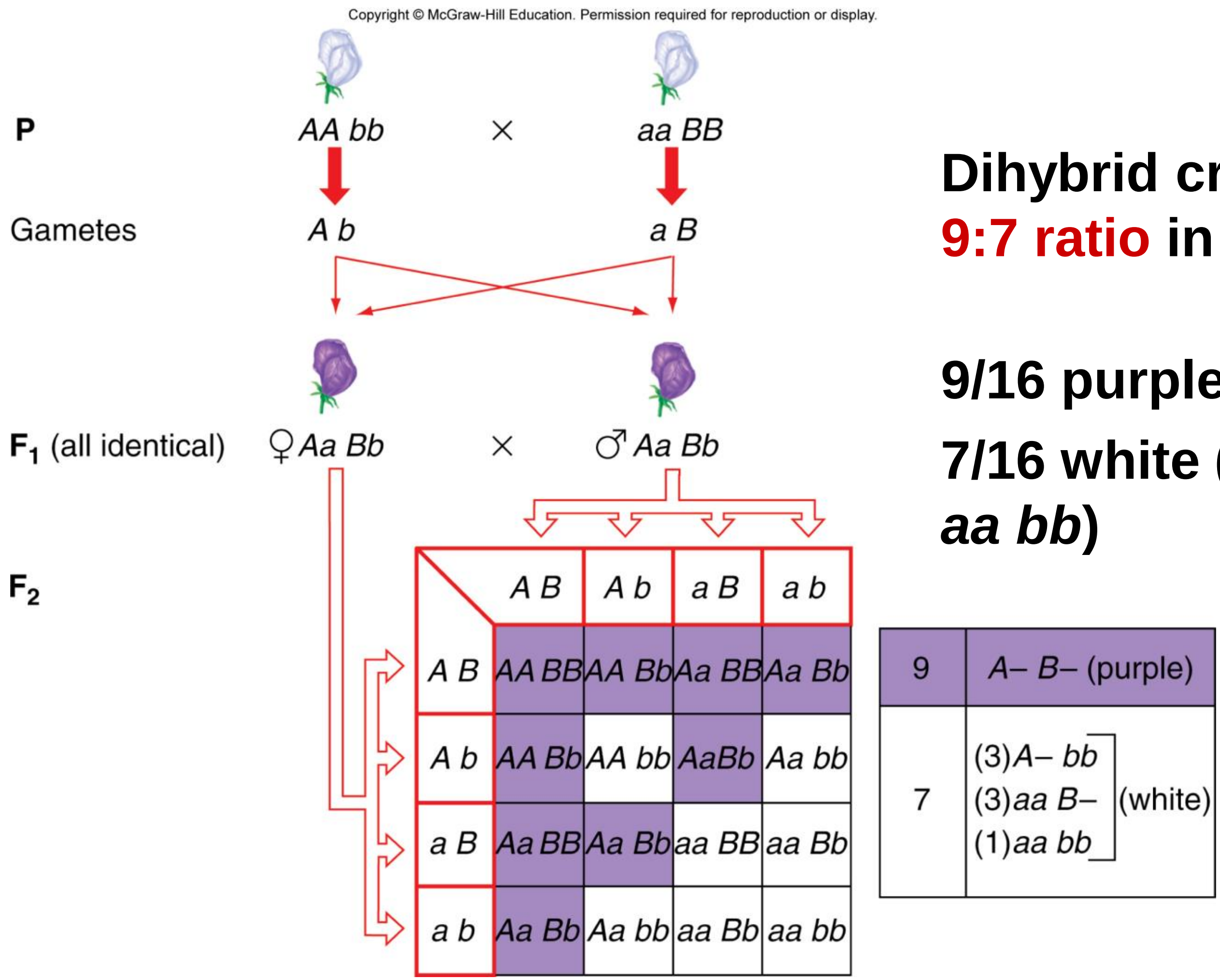
The A and B genes have not been identified, but genes for seed color in other plants suggest a model for determining lentil seed color.



Complementary gene action in sweet peas

Purple F_1 progeny are produced by crosses of two pure-breeding white lines

Complementary gene action generates purple flower color in sweet peas

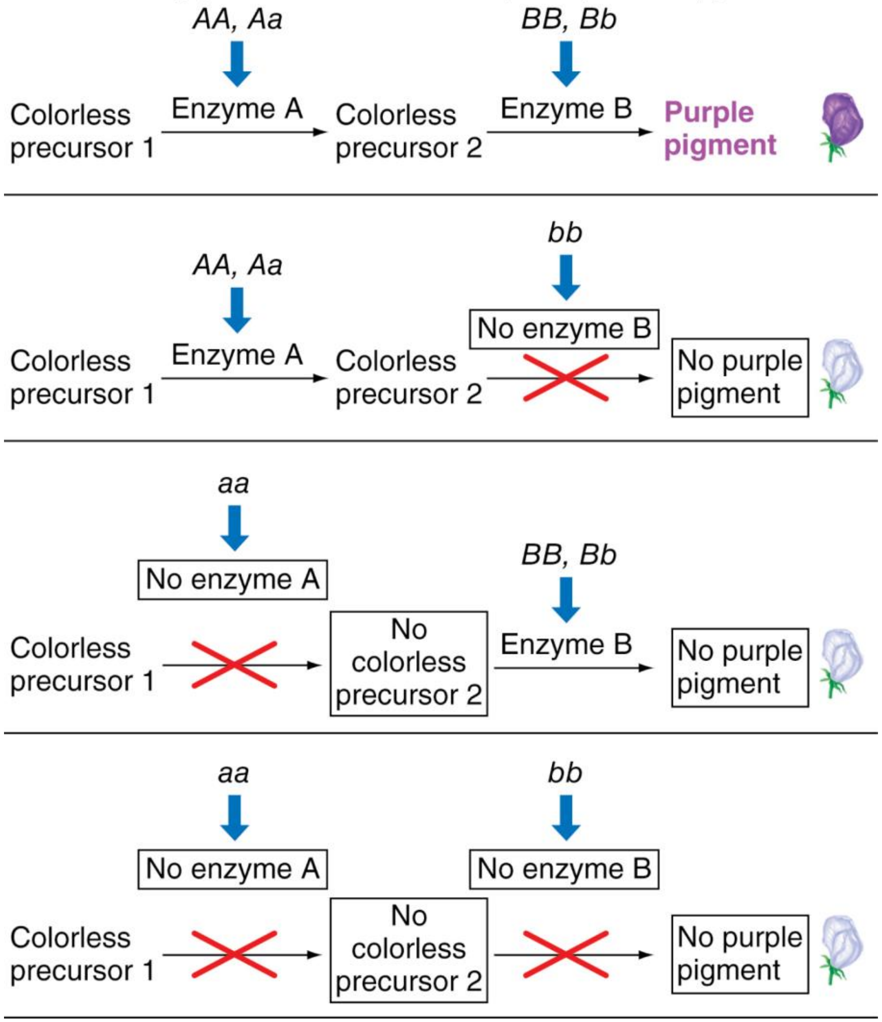


Dihybrid cross generates 9:7 ratio in F_2 progeny

9/16 purple (*A—B—*)

7/16 white (*A— bb*, *aa B—*, *aa bb*)

Possible biochemical explanation for complementary gene action for flower color in sweet peas



One pathway has two reactions catalyzed by different enzymes

- **At least one dominant allele of both genes is required for purple pigment**
- **Homozygous recessive for either or both genes results in no pigment**

Epistasis results from the effects of an allele at one gene masking the effects of another gene

The allele that does the masking is epistatic to the other gene.

The gene that is masked is hypostatic to the other allele.

Epistasis can be recessive or dominant

- **Recessive – epistatic allele must be homozygous recessive**
- **Dominant – One copy of an allele masks the other gene**

Coat color in Labrador retrievers is determined by two genes

Gene *B* alleles determine black and brown.

Recessive allele *ee* of gene *E* is epistatic to *B* and determines yellow.

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Recessive epistasis in Labrador retrievers

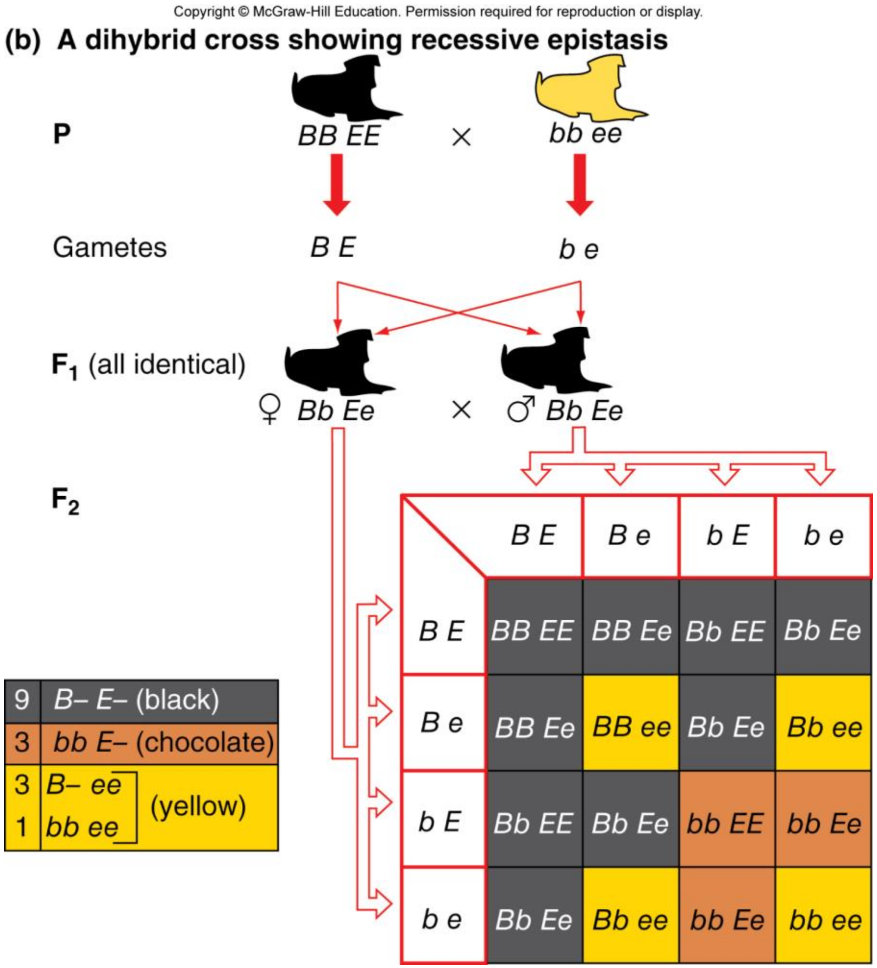
9:3:4 ratio in F_2 progeny of dihybrid crosses indicates recessive epistasis

9/16 black (*B*— *E*—)

3/16 brown (*bb E*—)

4/16 yellow (*B*— *ee*, *bb ee*)

Genotype *ee* masks the effect of all *B* genotypes



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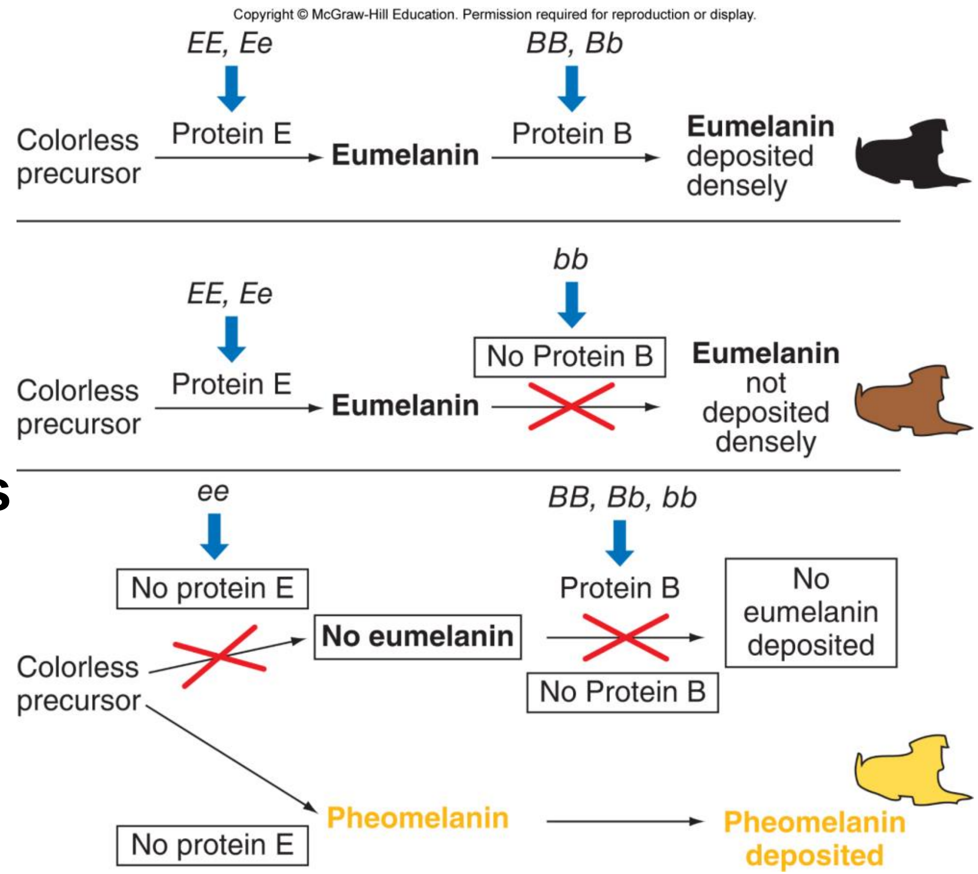
A biochemical explanation for coat color in Labrador retrievers

Protein B deposits eumelanin.

B– deposits eumelanin densely (black)

bb deposits eumelanin less densely (brown)

ee animals cannot make eumelanin (yellow)



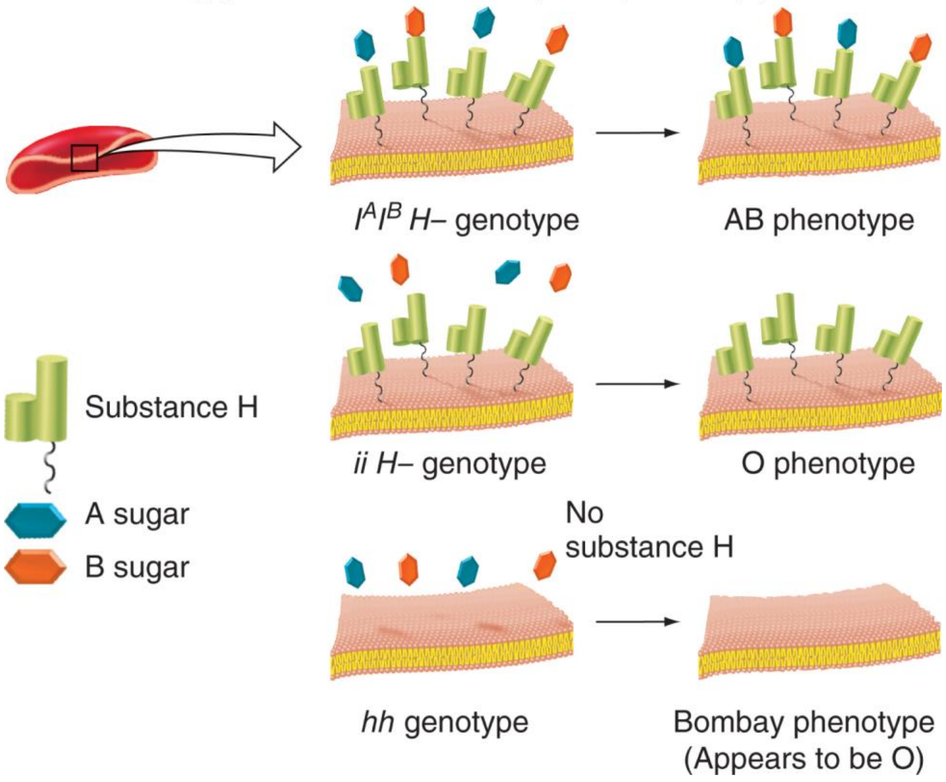
Recessive epistasis in humans with a rare blood type

Gene for substance H is epistatic to the ABO gene

- **Without the H substance, there is nothing for the A or B sugar to attach to**

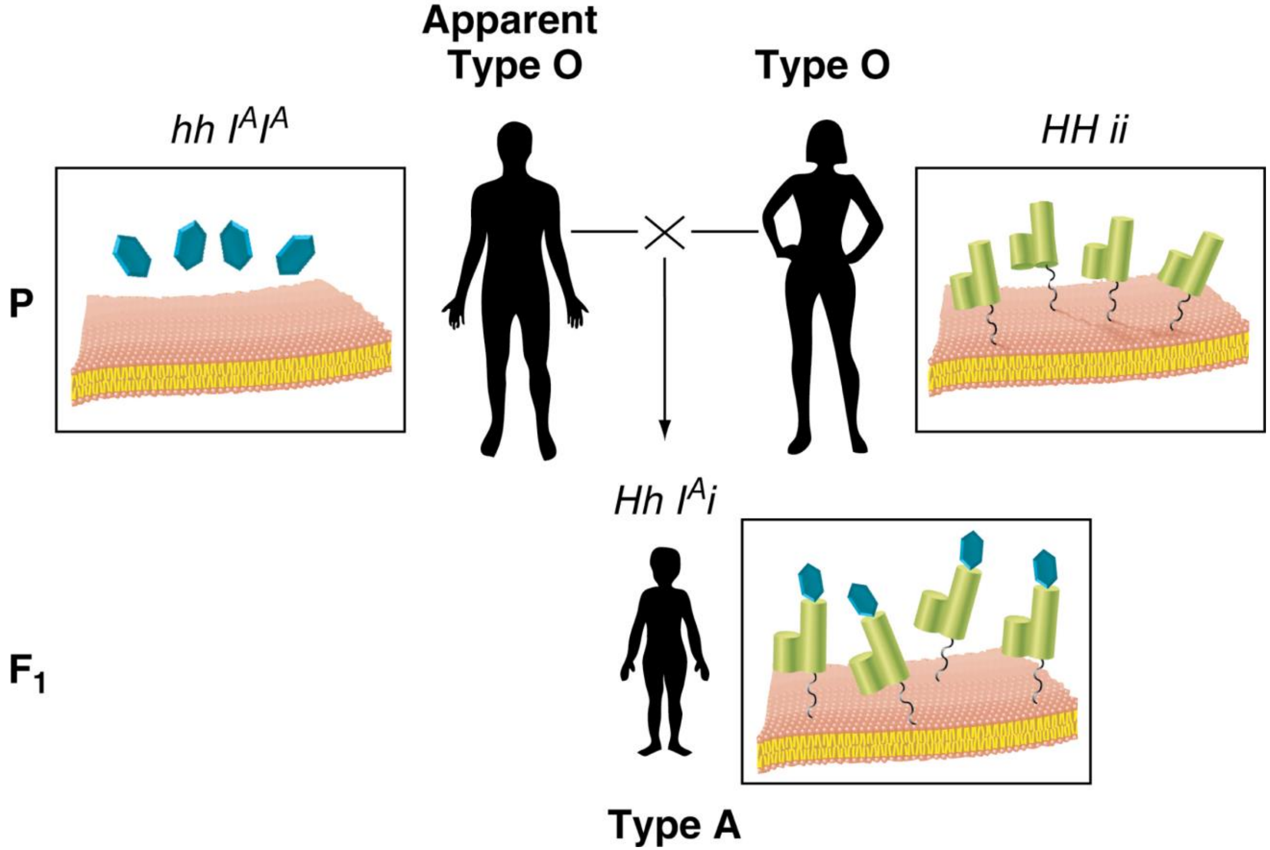
All type A, type AB, type B, and type O people are *H*—

People with *hh* genotype will <u>appear</u> to be type O



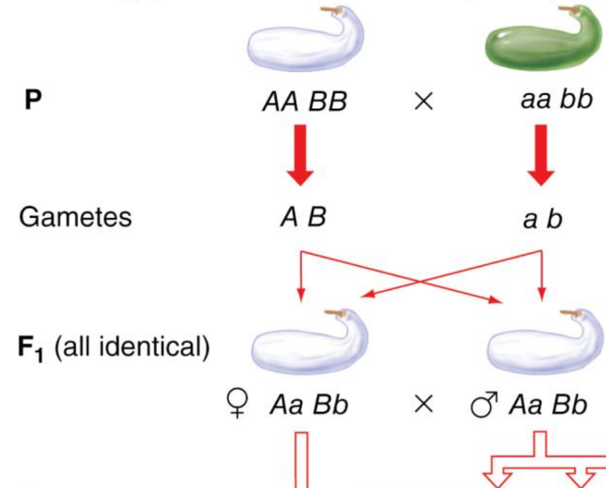
Epistasis causes unexpected inheritance pattern of ABO blood type

In this case, the male appears to have blood type O, but is able to pass an I^A allele to his child.



Dominant epistasis I in summer squash

12:3:1 ratio in F_2 progeny of dihybrid crosses indicates dominant epistasis I



	A B	A b	a B	a b
A B	AA BB	AA Bb	Aa BB	Aa Bb
A b	AA Bb	AA bb	Aa Bb	Aa bb
a B	Aa BB	Aa Bb	aa BB	aa Bb
a b	Aa Bb	Aa bb	aa Bb	aa bb

12	(9)*A– B–* (3)*aa B–*	(white)
3	*A– bb*	(yellow)
1	*aa bb*	(green)

12/16 white (*A— B—, aa B—*)

3/16 yellow (*A— bb*)

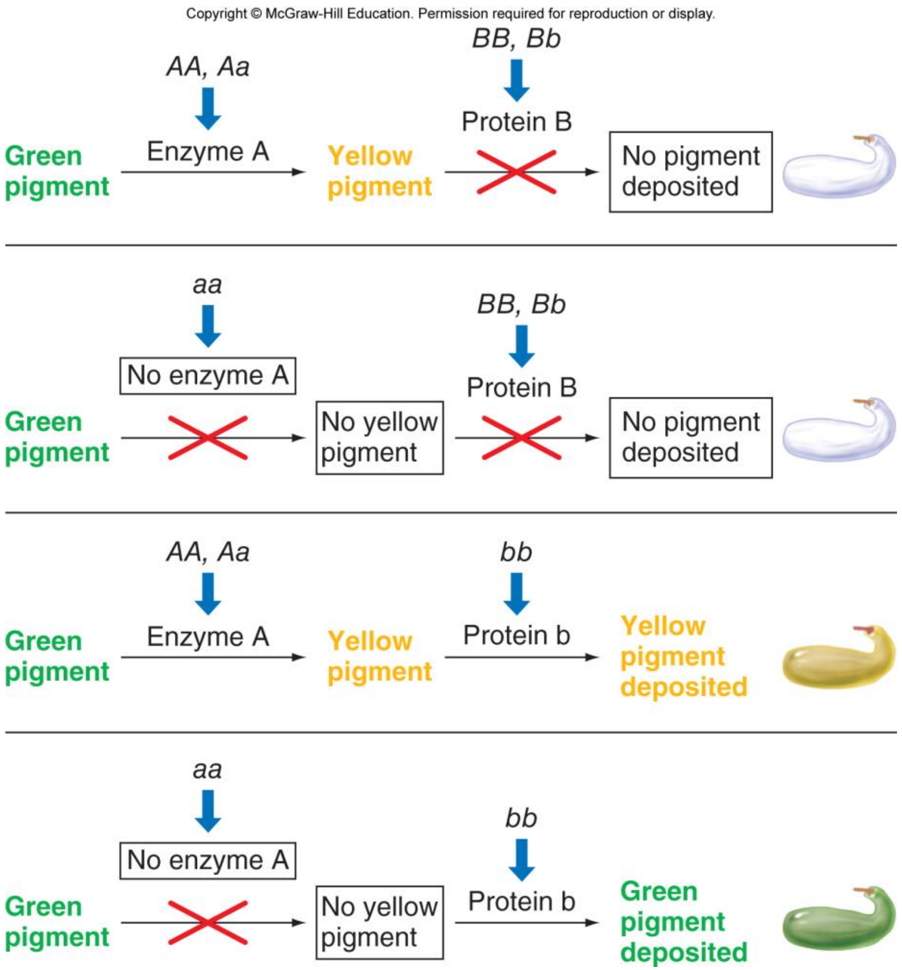
1/16 green (*aa bb*)

The dominant allele of one gene masks both alleles of another gene

A biochemical explanation for dominant epistasis in summer squash

Protein B is a dominant allele that prevents pigment deposition.

It is epistatic to any *A* allele.



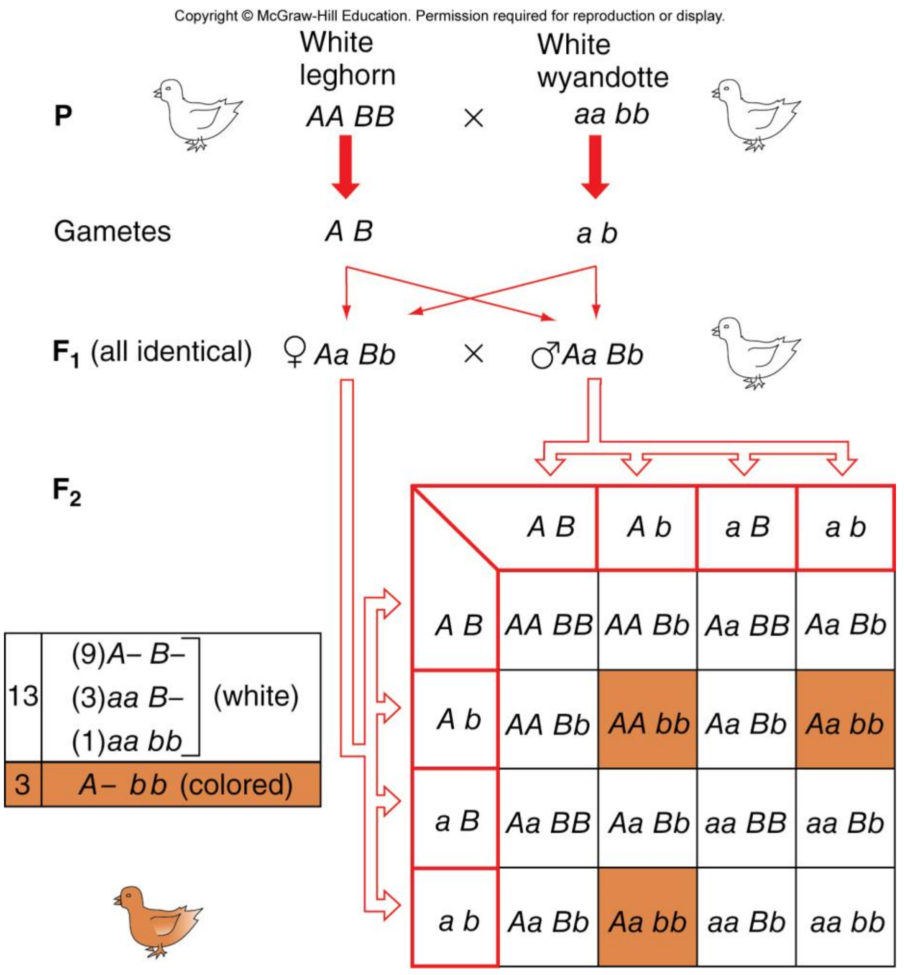
Dominant epistasis II in chicken feather color

13:3 ratio in F_2 progeny of dihybrid crosses indicates dominant epistasis II

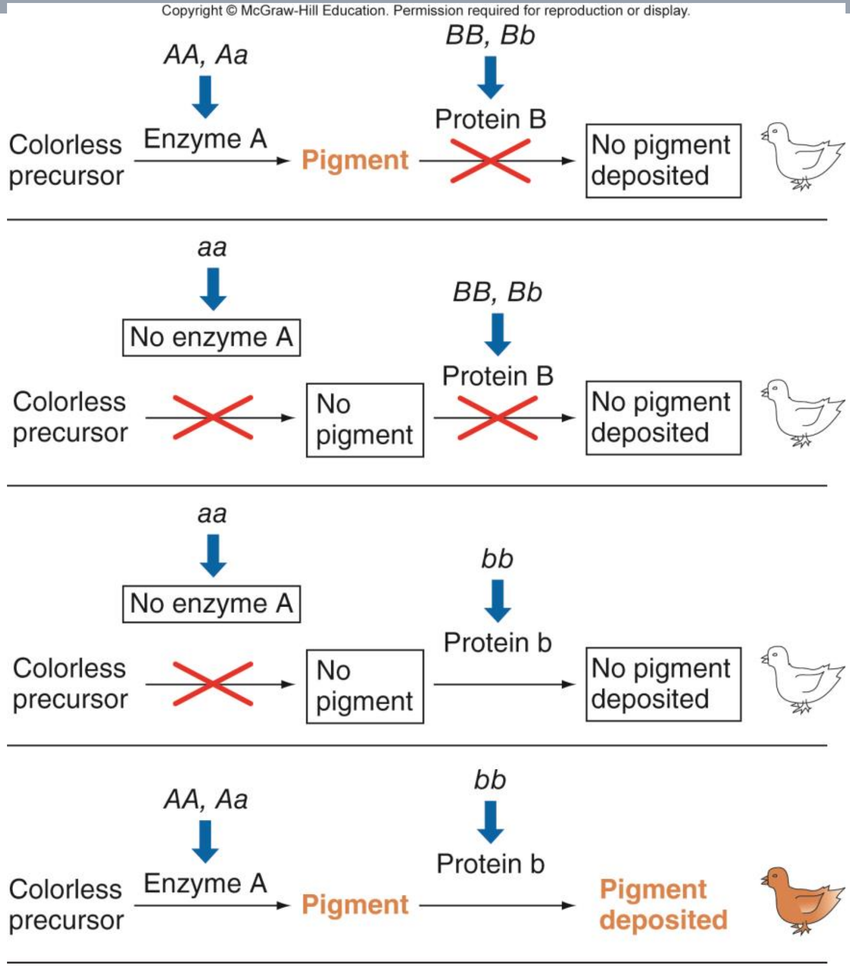
13/16 white (*A— B—*, *aa B—*, *aa bb*)

3/16 colored (*A— bb*)

The dominant allele of one gene masks the dominant allele of another gene



A molecular explanation for dominant epistasis in chicken feather color

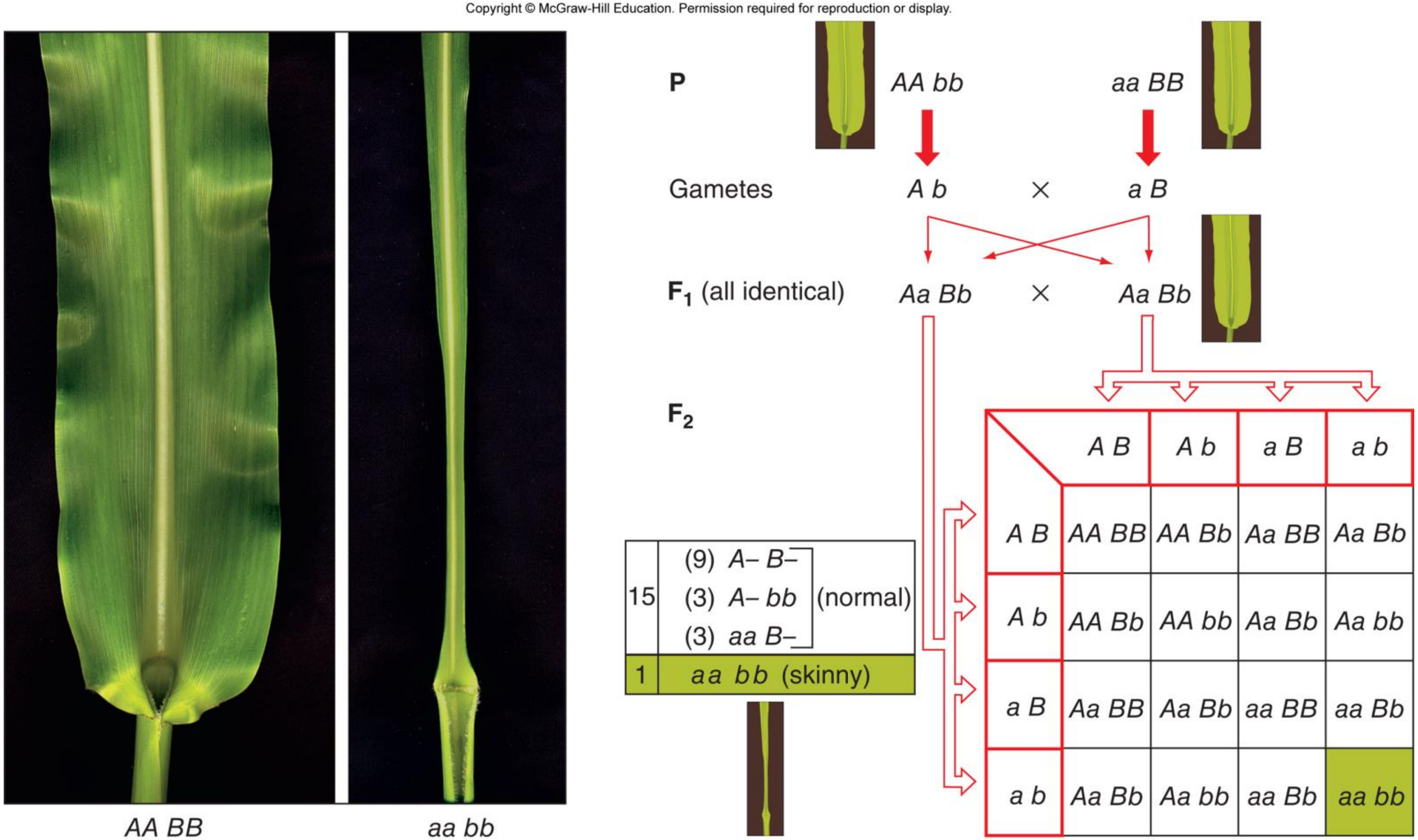


Note: The precursor in this biochemical pathway is colorless.

Allele *b* is epistatic to the *A* allele.

Redundant genes control leaf development in maize

Redundant genes result in the 15:1 phenotypic ratio



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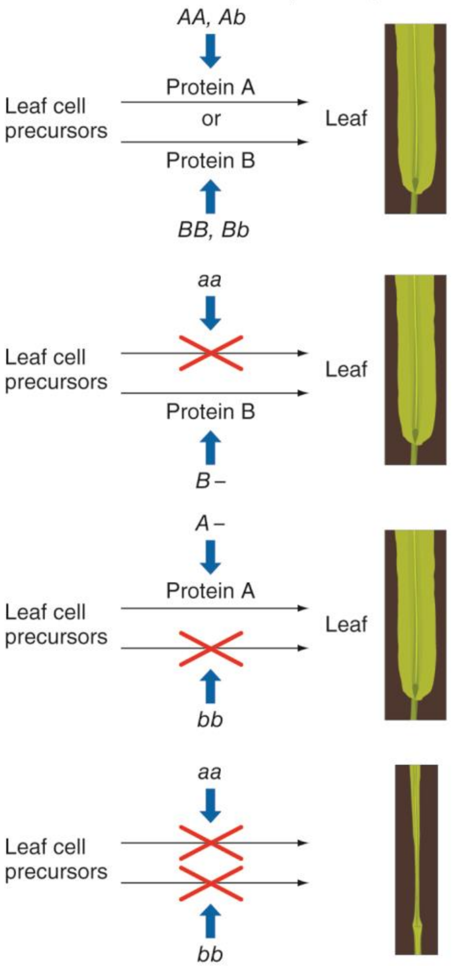
Proteins encoded by redundant genes perform nearly the same function

In this case, they recruit precursor cells to become part of the leaf.



Summary of gene interactions discussed in this chapter

Observing the F_2 ratios below is diagnostic of the type of gene interaction

- **These F_2 ratios occur only in dihybrid crosses where there is complete dominance**

TABLE 3.2 Summary of Gene Interactions

Gene Interaction	Example	F_2 Genotypic Ratios from an F_1 Dihybrid Cross: *A– B–*	*A– bb*	*aa B–*	*aa bb*	F_2 Phenotypic Ratio
None: Four distinct F_2 phenotypes	Lentil: seed coat color (see Fig. 3.10a)	9	3	3	1	9:3:3:1
Complementary: One dominant allele of each of two genes is necessary to produce phenotype	Sweet pea: flower color (see Fig. 3.12b)	9	3	3	1	9:7
Recessive epistasis: Homozygous recessive of one gene masks both alleles of another gene	Labrador retriever: coat color (see Fig. 3.14b)	9	3	3	1	9:3:4
Dominant epistasis I: Dominant allele of one gene hides effects of both alleles of another gene	Summer squash: color (see Fig. 3.17a)	9	3	3	1	12:3:1
Dominant epistasis II: Dominant allele of one gene hides effects of dominant allele of other gene	Chicken feathers: color (see Fig. 3.18a)	9	3	3	1	13:3
Redundancy: Only one dominant allele of either of two genes is necessary to produce phenotype	Maize: leaf development (see Fig. 3.19b)	9	3	3	1	15:1

Heterogeneous traits and the complementation test

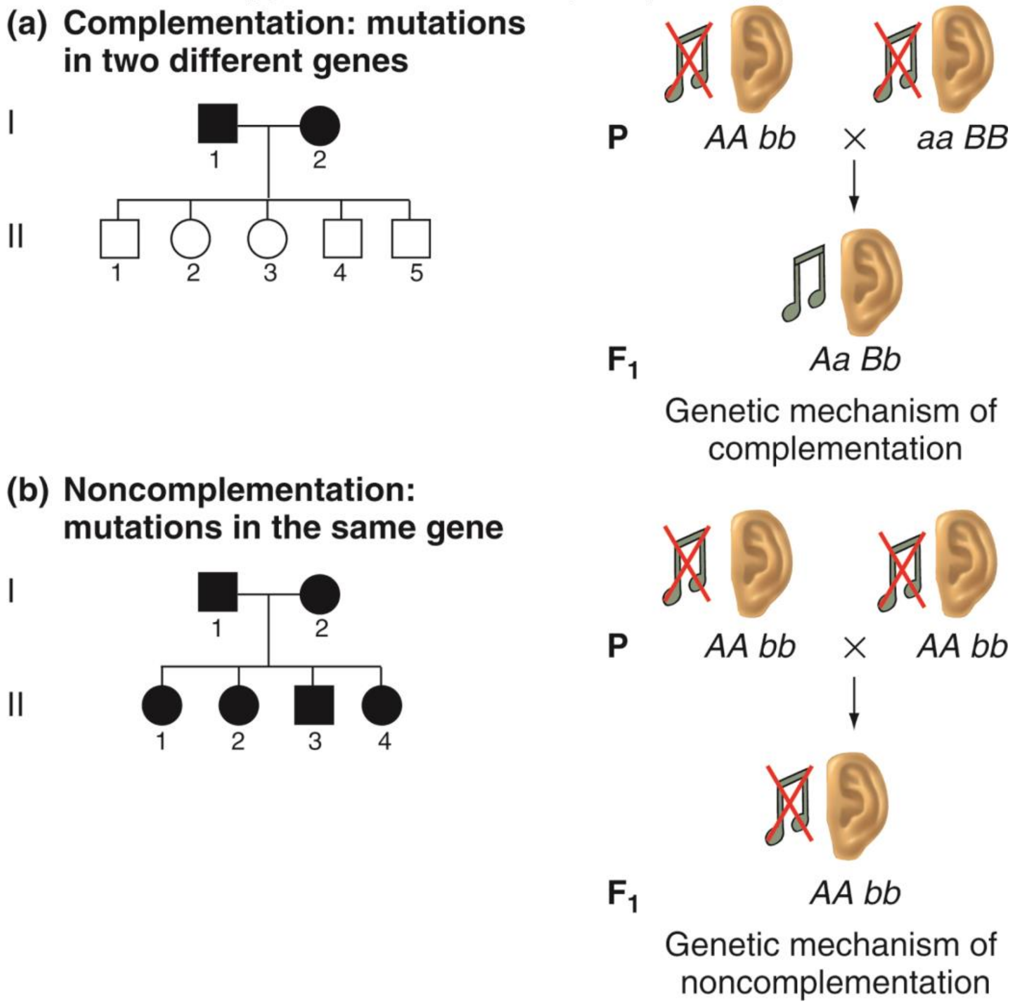
Heterogeneous traits have the same phenotype but are caused by mutations in different genes

- Deafness in humans can be caused by mutations in ~ 50 different genes

Complementation testing is used to determine if a particular phenotype arises from mutations in the same or separate genes

- Can be applied only with recessive, <u>not</u> dominant, phenotypes
- Discussed more in Chapter 7

Genetic heterogeneity in humans: Mutations in many genes can cause deafness



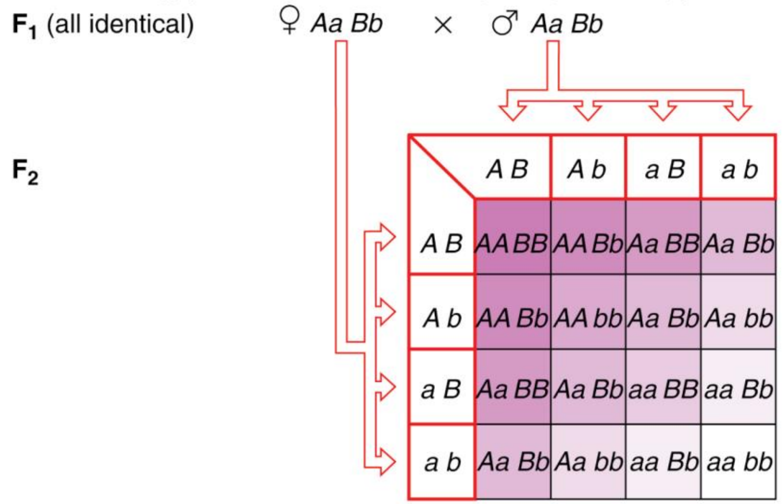
Interaction of two incompletely dominant genes can produce nine phenotypes

Example, two genes A and B:

- Allele *A* is incompletely dominant to allele *a*
- Allele *B* is incompletely dominant to allele *b*

For each gene, two alleles generate three phenotypes

- F_2 progeny have 3^2 phenotypes



1	*AA BB*	purple shade 9
2	*AA Bb*	purple shade 8
2	*Aa BB*	purple shade 7
1	*AA bb*	purple shade 6
4	*Aa Bb*	purple shade 5
1	*aa BB*	purple shade 4
2	*Aa bb*	purple shade 3
2	*aa Bb*	purple shade 2
1	*aa bb*	purple shade 1 (white)

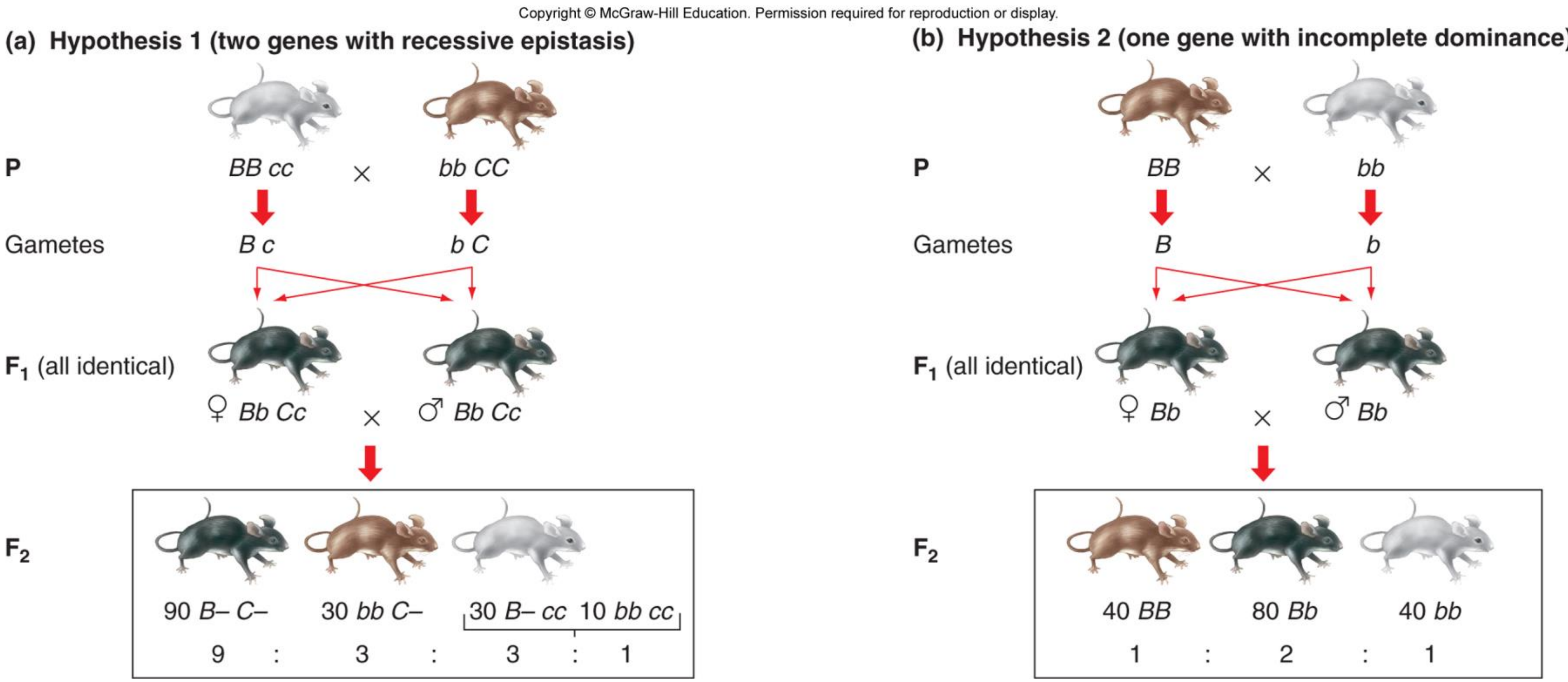
Breeding studies help determine inheritance of a trait

How do we know if a trait is caused by one gene or by two genes that interact?

Example: Dihybrid cross of pure-breeding parents produces three phenotypes in F_2 progeny

- **If single gene with incomplete dominance, then F_2 progeny should be in 1:2:1 ratio**
- **If two independently assorting genes and recessive epistasis, then F_2 progeny should be in 9:3:4 ratio**
- **These ratios can appear very similar. Further breeding studies can reveal which hypothesis is correct.**

Two hypotheses to explain phenotypes in F2 progeny of mice with different coat colors

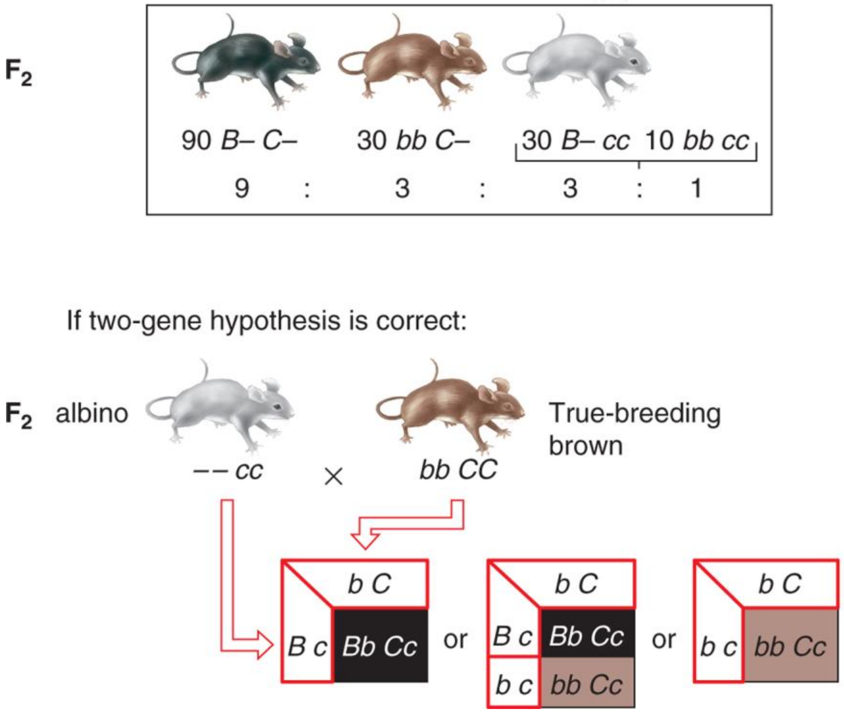


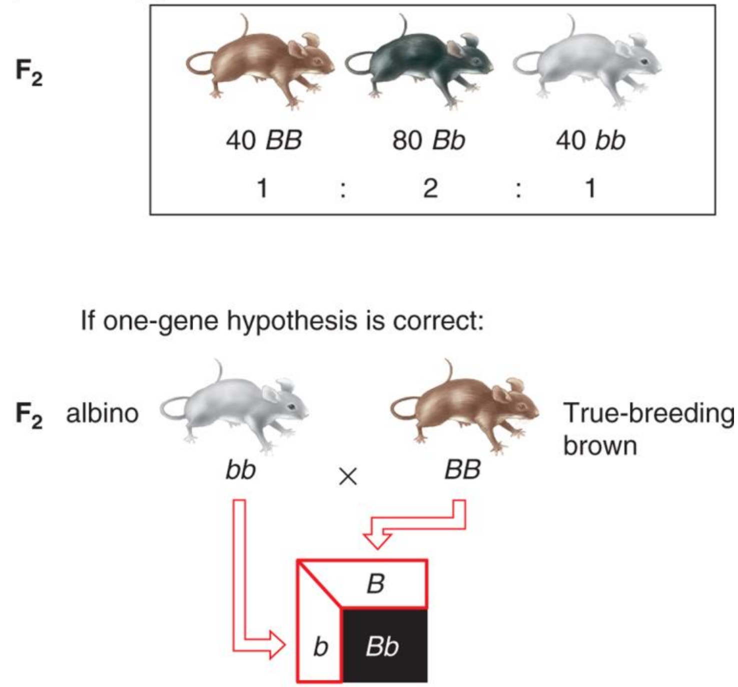
Are these F_2 progeny in a ratio of 9:3:4 or 1:2:1?

Specific breeding tests can help decide between two hypotheses

Hypothesis 1 – two genes with recessive epistasis

Hypothesis 2 – one gene with incomplete dominance



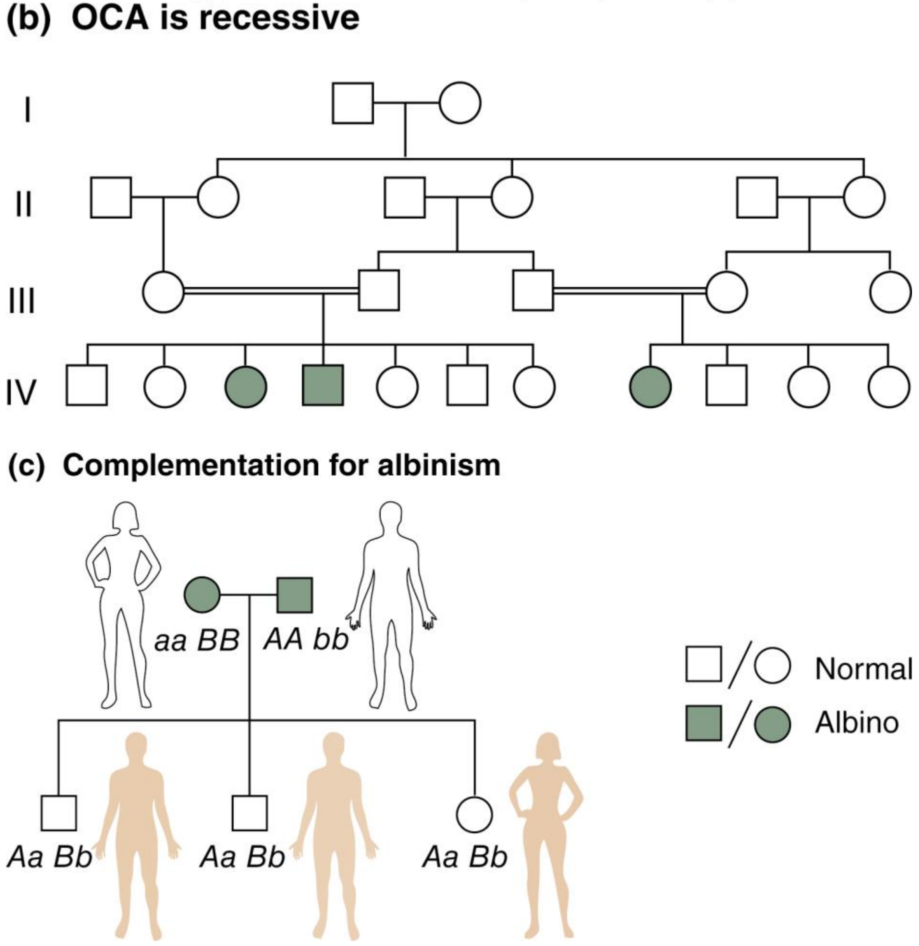


Family pedigrees help unravel the genetic basis of ocular-cutaneous albinism (OCA)

OCA is another example of heterogeneity

(a) Ocular-cutaneous albinism (OCA)

© Radu Sigheti/Reuters/Landov



The same genotype does not always produce the same phenotype

In all of the traits discussed so far, the relationship between a specific genotype and its corresponding phenotype has been absolute

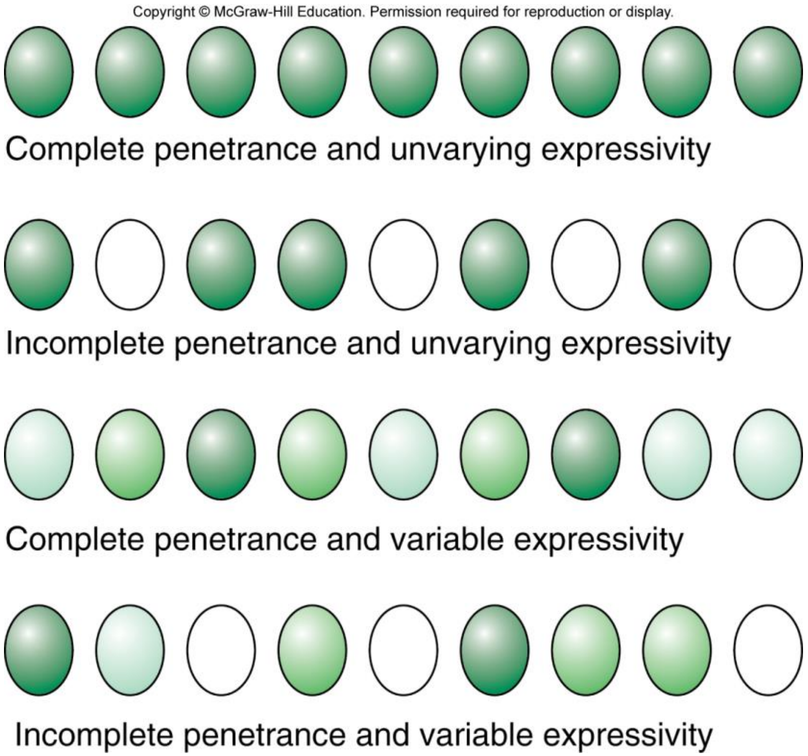
Phenotypic variation for some traits can occur because of:

- Effects of modifier genes
- Effects of environment
- Pure chance

Phenotype often depends on penetrance and/or expressivity

Penetrance is the percentage of individuals with a particular genotype that show the expected phenotype

- **Can be complete (100%) or incomplete**



Expressivity is the degree or intensity with which a particular genotype is expressed in a phenotype

- **Can be variable or unvarying**

Some traits result from different genes that do not contribute equally to the phenotype

Modifier genes alter the phenotypes produced by alleles of other genes

- Can have major effect or more subtle effects

Example: T locus of mice

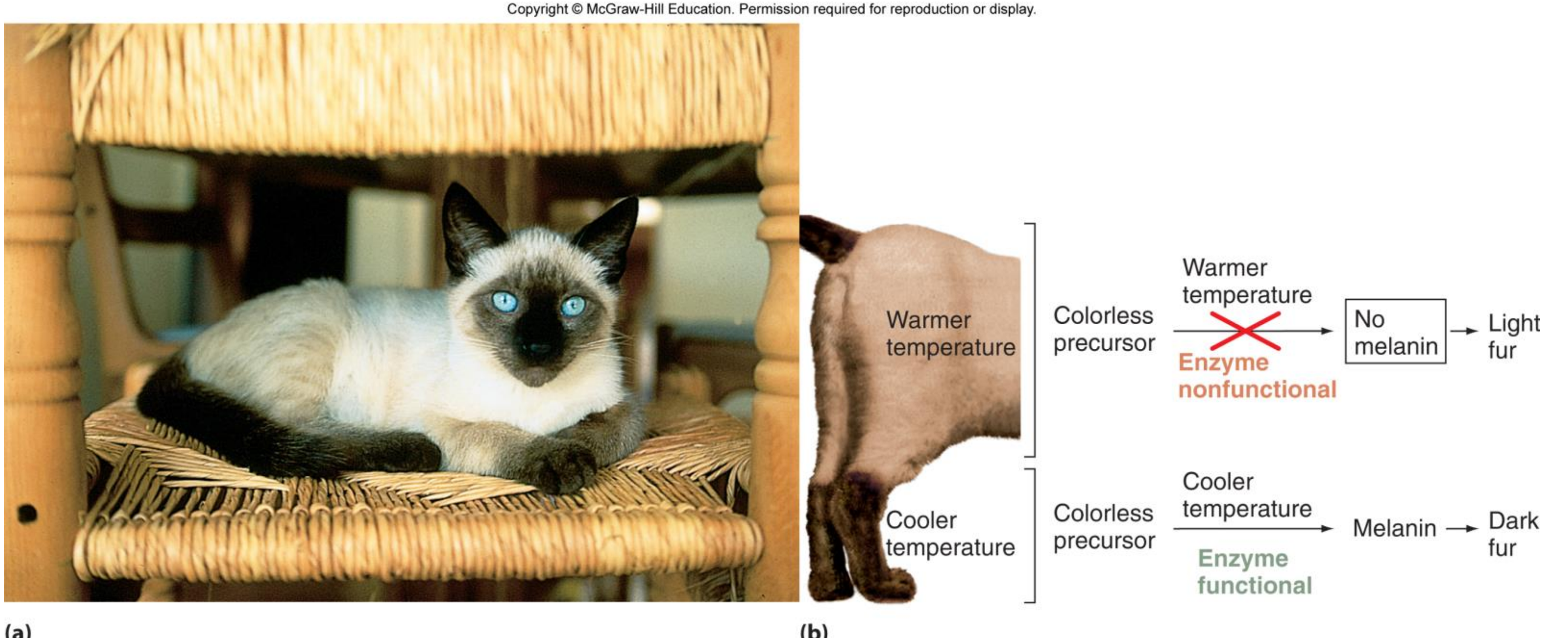
- Mutant *T* allele causes abnormally short tail.
- In different strains, the tails can be 75% of normal length or 10% of normal length.
- Different inbred strains must carry alternative alleles of a gene that modifies the *T* mutant phenotype.

Environmental effects on phenotype

Temperature is a common element of the environment that can affect phenotype

Coat color in Siamese cats

- Extremities are darker than body because of a temperature sensitive allele



Environmental effects on phenotype

Temperature affects survivability of a Drosophila mutant

- ***Shibire*** **mutants develop normally at < 29^{o}C but are inviable at temperatures > 29^{o}C**

Conditional lethal mutations are lethal only under some conditions

- **Permissive conditions - mutant allele has wild-type functions**
- **Restrictive conditions - mutant allele has defective functions**

Other effects of environment on phenotype

Phenocopy - phenotype arising from an environmental agent that mimics the effect of a mutant gene

- **Not heritable**
- **Can be deleterious or beneficial**
- **Examples in humans**
  - **Thalidomide produced a phenocopy of phocomelia, a rare dominant trait**
  - **Children with heritable PKU can receive a protective diet**
  - **Genetic predisposition to cardiovascular disease can be influenced by diet and exercise**
  - **Genetic predisposition to lung cancer is strongly affected by cigarette smoking**

Mendelian principles can also explain continuous variation

Discontinuous traits give clear-cut, "either-or" phenotypic differences between alternative alleles

- Example: All of the traits Mendel studied in peas were discontinuous

Continuous traits are determined by segregating alleles of many genes that interact together and with the environment

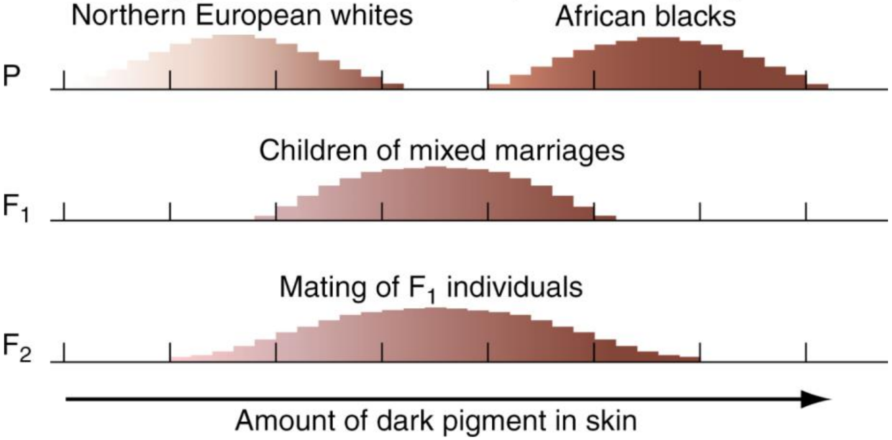
- Often appear to blend and "unblend"
- Also called **quantitative traits** because the traits vary over a range that can be measured
- Usually **polygenic** – controlled by multiple genes

Two continuous traits in human

Height is a continuous trait

© Rudi Von Briel/Photo Edit

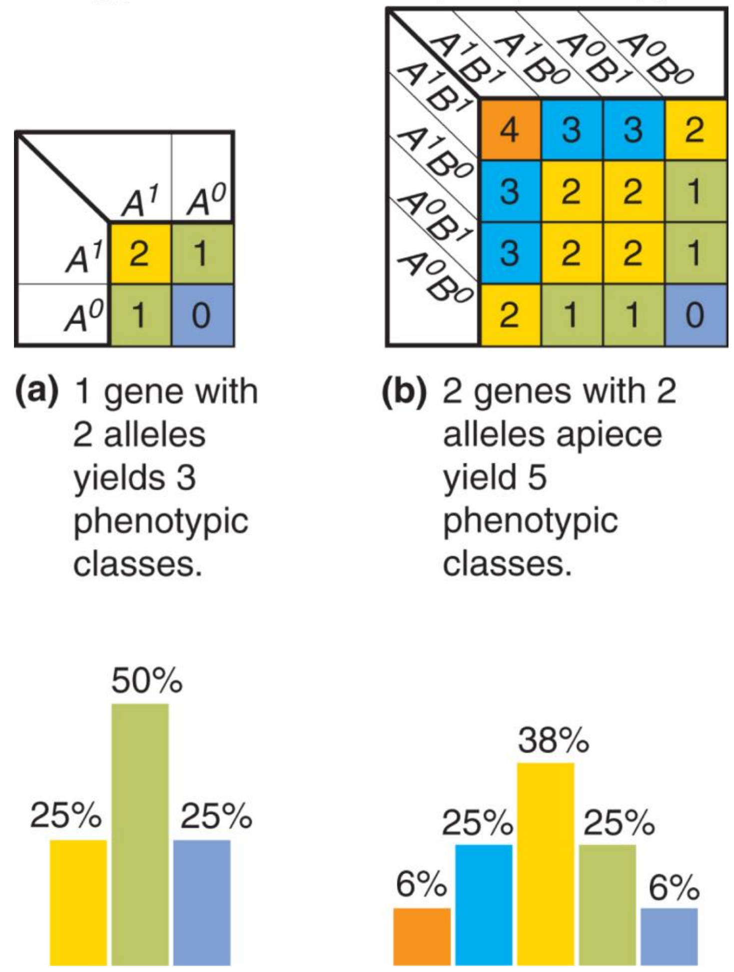
Skin color is a continuous trait



Mendelian explanation of continuous variation

More genes or alleles result in more phenotypic classes and more similarity to continuous variation.

In these examples, alleles are incompletely dominant and have additive effects



(a) 1 gene with 2 alleles yields 3 phenotypic classes.

(b) 2 genes with 2 alleles apiece yield 5 phenotypic classes.

Mendelian explanation of continuous variation (continued)

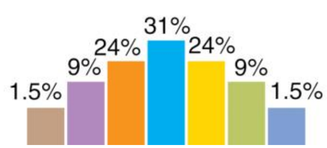
More genes or alleles result in more phenotypic classes and more similarity to continuous variation.

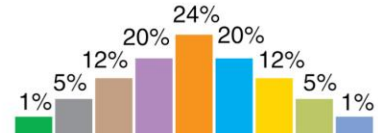
	$A^1B^1C^1$	$A^1B^0C^1$	$A^1B^1C^0$	$A^1B^0C^0$	$A^0B^1C^1$	$A^0B^0C^1$	$A^0B^1C^0$	$A^0B^0C^0$
$A^1B^1C^1$	6	5	5	4	5	4	4	3
$A^1B^0C^1$	5	4	4	3	4	3	3	2
$A^1B^1C^0$	5	4	4	3	4	3	3	2
$A^1B^0C^0$	4	3	3	2	3	2	2	1
$A^0B^1C^1$	5	4	4	3	4	3	3	2
$A^0B^0C^1$	4	3	3	2	3	2	2	1
$A^0B^1C^0$	4	3	3	2	3	2	2	1
$A^0B^0C^0$	3	2	2	1	2	1	1	0

(c) 3 genes with 2 alleles yield 7 phenotypic classes.

	A^2B^2	A^2B^1	A^1B^2	A^2B^0	A^0B^2	A^1B^1	A^1B^0	A^0B^1	A^0B^0
A^2B^2	8	7	7	6	6	6	5	5	4
A^2B^1	7	6	6	5	5	5	4	4	3
A^1B^2	7	6	6	5	5	5	4	4	3
A^2B^0	6	5	5	4	4	4	3	3	2
A^0B^2	6	5	5	4	4	4	3	3	2
A^1B^1	6	5	5	4	4	4	3	3	2
A^1B^0	5	4	4	3	3	3	2	2	1
A^0B^1	5	4	4	3	3	3	2	2	1
A^0B^0	4	3	3	2	2	2	1	1	0

(d) 2 genes with 3 alleles apiece yield 9 phenotypic classes.





A comprehensive example: Dog coat color is determined by multiple alleles of several genes

Genes E and B: Synthesis and deposition of the pigment eumelanin

- See Figure 3.15
- A third gene E allele (E^m) results in a dark mask

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A comprehensive example: Dog coat color is determined by multiple alleles of several genes

Genes *S* and *M*: Control spotting

- **Homozygotes for recessive allele of gene *S* (s^ps^p) are white with colored spots (due to other genes)**
- **The *M* gene has two codominant alleles.**
  - M^1 **phenotype is diluted color**
  - M^2 **phenotype is normal color**

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merle (M^1M^2)

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Some genes affecting domestic dog coat color and pattern

At least 14 genes are involved. Some have multiple alleles with incomplete dominance and epistasis.

TABLE 3.3 Some of the Genes Affecting Domestic Dog Coat Color and Pattern

Gene	Dominance Series of Alleles		Phenotypes
Gene E	$E^m > E > e$	E^m	Black mask on fawn or brindle
		E	Eumelanin (dark) and pheomelanin (yellow) pigments
		e	Only pheomelanin (cream, tan, red)
Gene B	$B > b$	B	Black: eumelanin deposited densely
		b	Brown: eumelanin deposited less densely
Gene A	$A^y > a^w > a^t > a$	A^y	Fawn (lots of light pigment on hair)
		a^w	Agouti (light stripe on dark hair)
		a^t	Tan belly (only hairs on belly have some light pigment)
		a	Black or brown (no light stripe on hairs)
Gene K	$K^b > k^{br} > k^y$	K^b	Solid color
		k^{br}	Brindled
		k^y	Gene A markings expressed normally
Gene D	$D > d$	D	Colors not dilute
		d	Colors dilute
Gene S	$S > s^p$	S	No white markings
		s^p	Colored patches on white background
Gene M	$M^1 = M^2$	M^1	Coat color diluted (homozygote has various health problems)
		M^2	Normal color